

ROAD HEALTH & CITIZEN GRIEVANCE REPORT

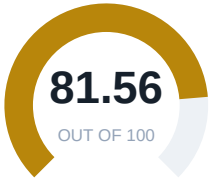
Automated computer-vision assessment of road surface condition, defect extent and remediation urgency, prepared for municipal grievance submission.

CONDITION: FAIR

RISK: HIGH · Priority-2

REF: RHA-20260907-FA6D20

ROAD HEALTH INDEX



FAIR

DEFECTS DETECTED

5

5 potholes

DEFECT AREA RATIO

18.44%

of the photographed surface

HIGHEST SEVERITY

Critical

Pothole - tag #1

ROAD RISK ASSESSMENT

Project-defined model - see methodology

ROAD RISK INDEX

54.7

HIGH RISK

Priority-2

Remediate within 7 days

REMEDIATION URGENCY

54.7 / 100



Extent		74 / 100	18.44% of the frame damaged weight 50%
Worst defect		100 / 100	10.86% of the frame in one box weight 30%
Density		62 / 100	5 defects detected weight 20%

Weighted mean of the components above, scaled by a hazard factor of 1.00 (how dangerous the detected classes are, area-weighted) and a confidence factor of 0.69 (mean confidence, floored at 0.50).

GEOSPATIAL & CAPTURE TELEMETRY

Date & time	07 Sep 2026, 08:57 AM
Coordinates	17.494800° N, 78.399600° E
Region	Hyderabad, Telangana, India
Source image	11_jpg.rf.7ef3d543d83c0f883b9145bb22c860a5.jpg (640 x 640 px)
Evidence SHA-256	50f360a7e64b3130f1a20b33...
Detector	best.pt (yolo11n-935e809b)
Map link	Open this location in Google Maps

MUNICIPAL JURISDICTION & ROUTING

Recipient body	Greater Hyderabad Municipal Corporation (GHMC)
Grievance tier	Priority-2 - Remediate within 7 days
Dominant defect	Pothole
Priority action	Barricade, excavate and full-depth patch with hot-mix asphalt
Submission status	Prepared by the citizen - not yet filed with the authority

Official Citizen Grievance Statement

Road surface defects were detected in a photograph captured at latitude 17.494800, longitude 78.399600. Automated analysis identified 5 potholes, covering approximately 18.44% of the photographed road surface. The computed Road Health Score is 81.56 out of 100, indicating a FAIR road condition. The estimated Road Risk Index is 54.7 out of 100, placing this location in the HIGH risk band; the suggested grievance tier is Priority-2 (remediate within 7 days). The most serious individual defect is rated critical severity. The Greater Hyderabad Municipal Corporation (GHMC) is kindly requested to inspect the location and undertake the necessary repair work.

Addressed to: Greater Hyderabad Municipal Corporation (GHMC)

DEFECT INVENTORY & RECOMMENDED REMEDIATION

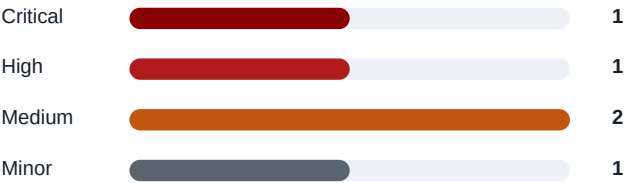
Suggestions for the engineering wing, not directives

TAG	CLASSIFICATION	DETECTION CONFIDENCE		SEVERITY	FRAME AREA	RECOMMENDED REMEDIATION
#1	Pothole		82%	Critical	10.86%	Barricade, excavate and full-depth patch with hot-mix asphalt
#2	Pothole		78%	High	3.46%	Hot-mix / cold-mix asphalt fill with mechanical compaction
#3	Pothole		66%	Medium	1.46%	Edge trimming, tack coat and patch sealing
#4	Pothole		63%	Medium	2.04%	Edge trimming, tack coat and patch sealing
#5	Pothole		55%	Minor	0.62%	Localised surface levelling during the next maintenance round

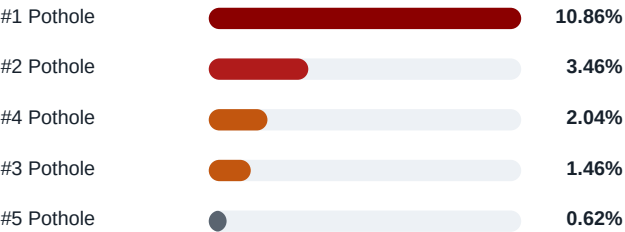
RISK ANALYTICS

How the index was arrived at

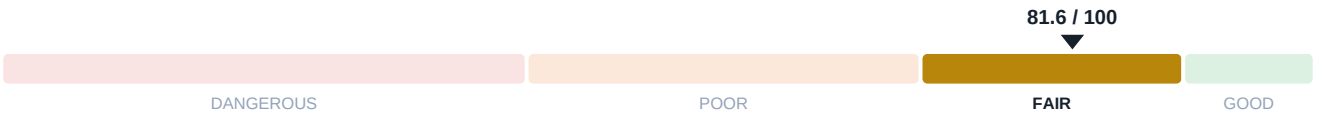
DEFECT COUNT BY SEVERITY BAND



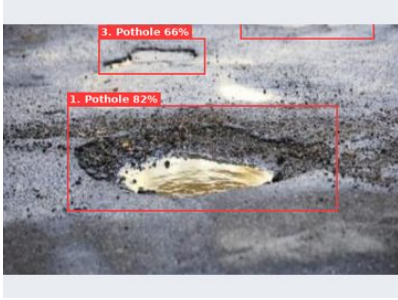
FRAME SHARE PER DEFECT



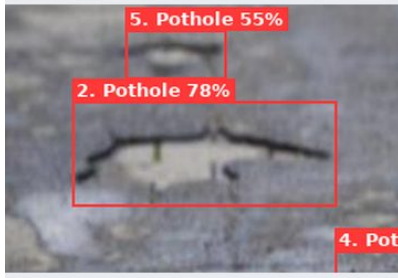
ROAD HEALTH SCORE AGAINST THE PROJECT'S CONDITION BANDS



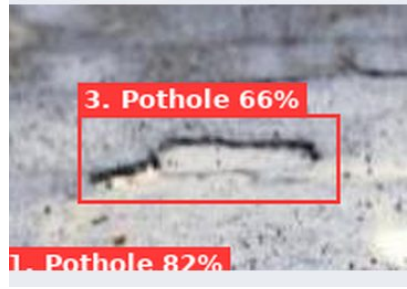
Health score = 100 - defect percentage = 100 - 18.44 = 81.56. Higher is better here, so this scale runs in the opposite direction to the risk scale on page 1.



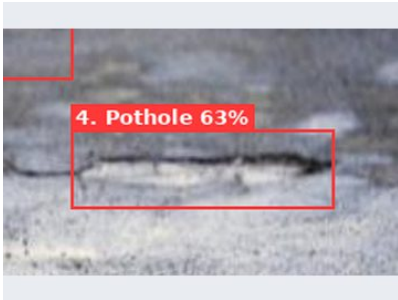
Tag #1 · Pothole
Critical · 82% confidence



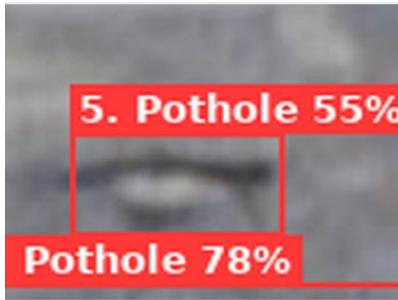
Tag #2 · Pothole
High · 78% confidence



Tag #3 · Pothole
Medium · 66% confidence



Tag #4 · Pothole
Medium · 63% confidence



Tag #5 · Pothole
Minor · 55% confidence



Figure 1. Detected defects with class and confidence. Each label number matches a tag in the defect inventory. 640 x 640 px.

Automated visual estimate from a single photograph. The score, severity and risk thresholds used in this report are defined by this project and are not taken from IRC, MoRTH, GHMC or any other official standard. A photograph carries no depth or scale, so severity reflects the share of the frame a defect occupies, not how deep it is. This report does not replace physical inspection by a qualified engineer.